Part D: Analysis, Validation and Engineering Decision

Aim and method

The aim is to analyse the reliability of the FHSS link under partial-band interference, study receiver hop-rate mismatch, compare FHSS with fixed-frequency BPSK, and recommend a suitable hop-set size and hop rate.

The channel model is the same as Part B: eight hopping channels, BPSK at 10 ksymbol/s, a sample rate of 500 ksample/s, and a 320-bit frame. Aggregate jammer-to-signal power ratio is fixed at $J/S = 10$ dB. Jammer occupancies are 12.5%, 25% and 50%, with no jammer as a reference. The three hop rates are 625, 1,250 and 2,500 hops/s.

The results below use the same recorded simulation run as the supplied Part B report. Thus, the numerical values and conclusions can be read as a continuation of Part B.

The two reliability measures are

$$\mathrm{BER} = \frac{\text{incorrect payload bits}}{\text{total transmitted payload bits}}, \qquad \mathrm{PER} = \frac{\text{failed packets}}{\text{total transmitted packets}}.$$

A packet fails if synchronization or CRC validation fails. CRC detects corruption; it does not correct the damaged bits.

1. Effect of jammer occupancy

If K of the $N = 8$ channels are jammed, the occupancy is

$$\rho = \frac{K}{8}.$$

Since total jammer power is constant and desired signal power is normalized to one,

$$J_{\text{total}} = 10, \qquad J_{\text{channel}} = \frac{10}{K}.$$

Occupancy	Jammed channels	Jammer power per occupied channel
12.5%	1	10
25%	2	5
50%	4	2.5

A narrow jammer causes fewer but stronger hop collisions. A wider jammer affects more hops but has less power on each occupied channel. Therefore, doubling occupancy does not necessarily double BER.

At 1,250 hops/s and $E_b/N_0 = 10$ dB, the measured results are:

Jammer occupancy	Payload BER	PER
0%	3.906×10^{-6}	0.0025
12.5%	0.02744	1
25%	0.05011	1
50%	0.07344	1

BER increases as occupancy increases in this experiment. At 25% occupancy, about 5% of payload bits are incorrect, even though most individual bits are still recovered correctly.

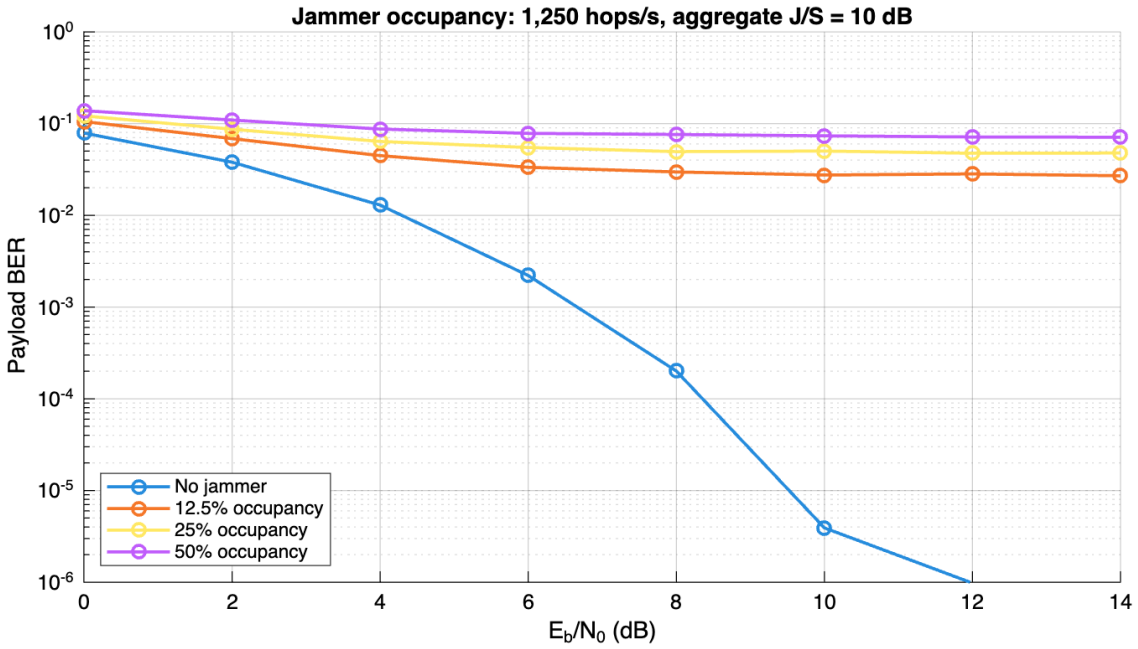


Figure D1. Payload BER against E_b/N_0 at 1,250 hops/s for four jammer occupancies. Aggregate $J/S = 10$ dB for the jammer cases.

Without a jammer, BER decreases rapidly as E_b/N_0 increases. The measured curve agrees with the expected behaviour of coherent BPSK in AWGN.

With a jammer, the curves flatten at high E_b/N_0. For example, at 25% occupancy, BER is approximately 0.0638 at 4 dB, 0.0501 at 10 dB and 0.0477 at 14 dB. Increasing E_b/N_0 reduces ordinary noise errors, but the jammer continues to corrupt the occupied hops. This residual BER is called an error floor.

PER is more severe than BER. All 250 packets failed at each nonzero-jammer operating point in the main sweep. One detected error in a CRC-protected field can reject the entire packet. Hence, a BER of only a few percent can still produce almost complete packet loss.

An observed PER of 1 is a finite simulation result, not proof that the true error probability is exactly one. For 250 failures in 250 trials, the 95% Wilson interval is approximately 0.985 to 1.

The main occupancy sweep supplies the correct packet timing to the receiver. Therefore, this graph measures the aligned link; unknown-timing acquisition is considered separately.

2. Effect of hop-rate mismatch

The transmitter and receiver must change channels at matching times. Let the nominal dwell time be T_d. In the mismatch experiment, receiver dwell time is

$$T_{d,\mathrm{rx}} = T_d(1+\epsilon),$$

where ϵ is the fractional dwell-duration error. A positive error makes the receiver dwell longer and hop more slowly. A negative error makes it hop faster.

After H hops, the approximate boundary drift is

$$\Delta t \approx H\epsilon T_d.$$

Even a small error accumulates across a frame. The receiver eventually listens on a different frequency from the transmitter, causing incorrect BPSK decisions.

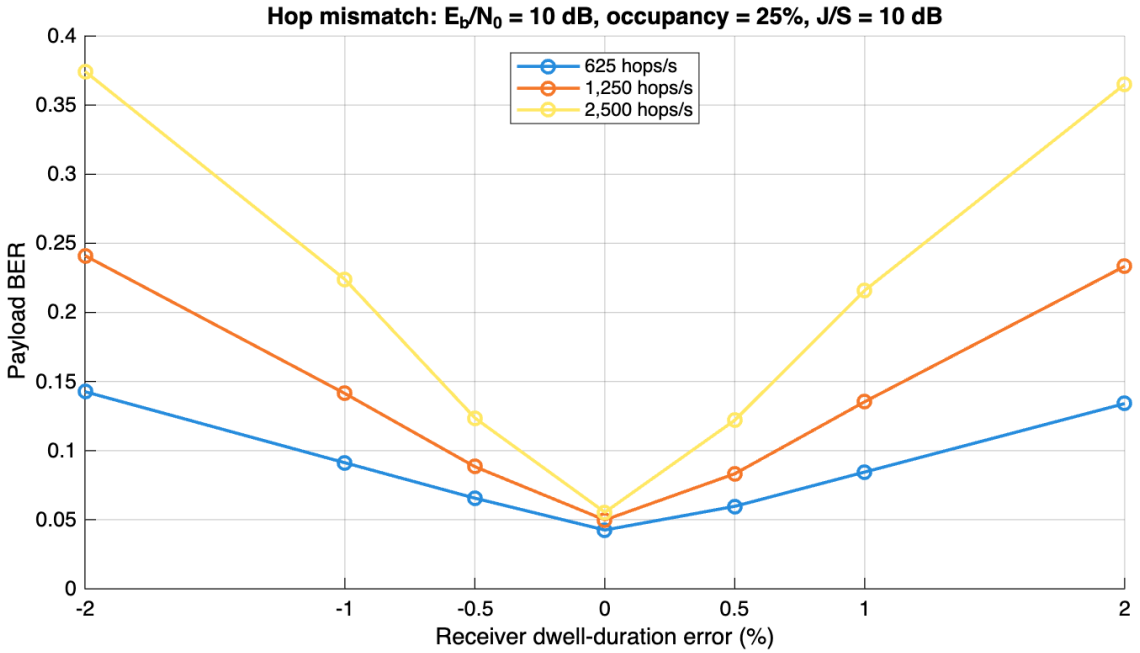


Figure D2. BER against receiver dwell-duration error for three hop rates, at $E_b/N_0 = 10$ dB, 25% occupancy and $J/S = 10$ dB.

Dwell error	BER at 625 hops/s	BER at 1,250 hops/s	BER at 2,500 hops/s
-2%	0.1426	0.2410	0.3744
-1%	0.0912	0.1413	0.2238
0%	0.0425	0.0498	0.0553
+1%	0.0845	0.1354	0.2159
+2%	0.1340	0.2334	0.3651

Each curve has its minimum near zero mismatch. Faster hopping is more sensitive to the same fractional dwell error because the accumulated drift covers a larger fraction of a short dwell.

For example, a frame lasts 32 ms. At 1% dwell error, boundary drift is approximately 0.32 ms. This is about 20%, 40% and 80% of one dwell at 625, 1,250 and 2,500 hops/s respectively. The calculation explains why the fastest rate shows the largest BER increase.

PER remains 1 throughout this mismatch study because the existing jammer already causes every tested packet to fail. Therefore, BER is the more useful measure for showing the additional damage caused by mismatch. The results do not show a further PER increase above one.

A wrong hopping-code position also causes severe degradation. A forced one-hop offset gives BER values of approximately 0.503, 0.498 and 0.487 at the three rates, close to random binary decisions. Correct hopping-code alignment is therefore essential.

3. FHSS and fixed-frequency comparison

The comparison uses the same BPSK symbol rate and average transmitted power. Four cases are studied: FHSS, a fixed carrier inside the jammed band, a fixed carrier outside it, and a fixed channel chosen uniformly at random for each packet.

This comparison uses the idealized symbol model with perfect channel isolation. It supports a relative comparison but should not be numerically mixed with the waveform results in Figures D1 and D2.

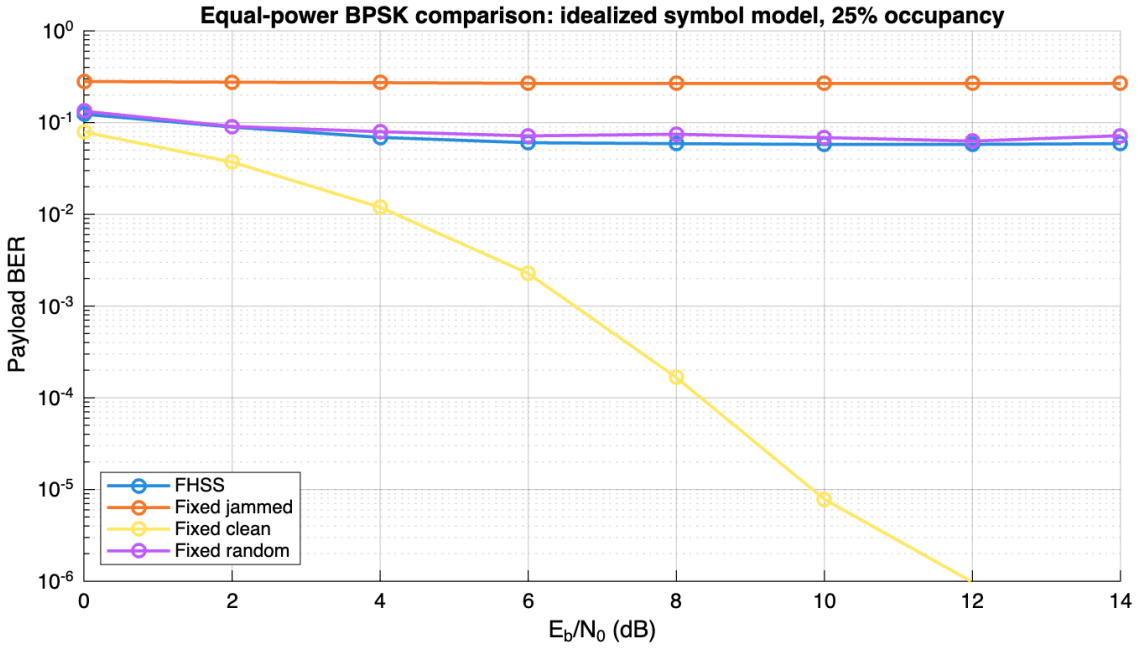


Figure D3. Equal-power BPSK comparison using the idealized symbol model. Jammer occupancy is 25%, aggregate $J/S = 10$ dB, and FHSS uses 1,250 hops/s.

At $E_b/N_0 = 10$ dB:

Link	Payload BER	PER
FHSS	0.05769	1
Fixed jammed carrier	0.26614	1
Fixed clean carrier	7.813×10^{-6}	0.002
Random fixed channel per packet	0.06850	0.2591

The fixed jammed link performs worst in BER because it remains inside the interference continuously. FHSS leaves the jammed channels and sends part of the message through clear channels. Its average BER is therefore much lower than that of the permanently jammed fixed link.

A guaranteed clean fixed channel gives the best result because it avoids the jammer completely. However, this assumes that a clean channel is known and remains available.

The random fixed-channel link has similar average BER to FHSS but much lower PER in this experiment. At 25% occupancy, approximately one quarter of packets choose a jammed carrier, while the others use clean carriers. FHSS distributes jammed symbols across almost every frame, so CRC can reject every frame even when its average BER is lower.

Thus, FHSS improves resistance to remaining on one jammed frequency, but does not guarantee lower PER than every fixed-frequency strategy. Without error correction or retransmission, the present jammer is too strong for reliable delivery of complete FHSS packets.

4. Validation of the results

Without interference, coherent BPSK should follow

$$P_b = \frac{1}{2}\operatorname{erfc}\left(\sqrt{E_b/N_0}\right).$$

At 4 dB and 1,250 hops/s, the measured BER is approximately 0.0129 against a theoretical value of 0.0125. At 10 dB, two payload errors in 512,000 tested bits give 3.906×10^{-6}, close to the theoretical 3.872×10^{-6}. These values support the AWGN scaling used in Part B.

Noiseless synchronized tests recover the exact payload at every hop rate. CRC-16 produces the reference value `0x29B1` for `123456789`. Seeded hopping is repeatable and visits all eight channels once in each complete permutation block.

5. Engineering recommendation

Hop-set size and bandwidth

The symbol rate is $R_s = 10$ ksymbol/s. Rectangular BPSK has an approximate first-null bandwidth of

$$B_{\text{ch}} \approx 2R_s = 20 \text{ kHz}.$$

For channel spacing $\Delta f = 25$ kHz and available hopping span 195 kHz,

$$B_{\text{hop}} = (N-1)\Delta f + B_{\text{ch}}.$$

Therefore,

$$N \leq 1 + \frac{195 - 20}{25} = 8.$$

The recommended hop set is eight channels. This uses the available main-lobe bandwidth and spreads transmission across the whole assigned band. The 195 kHz value is a first-null span estimate, not a strict limit on all radiated energy; rectangular pulses have sidelobes.

Hop rate

With eight symbols per hop,

$$T_d = \frac{8}{10,000} = 0.8 \text{ ms}, \qquad R_h = \frac{1}{T_d} = 1,250 \text{ hops/s}.$$

The recorded acquisition success probabilities at 10 dB and 25% occupancy are:

Hop rate	Acquisition success	Dwell time
625 hops/s	50%	1.6 ms
1,250 hops/s	92%	0.8 ms
2,500 hops/s	90%	0.4 ms

Each value is based on 50 acquisition trials. The two faster rates are within five percentage points of the best observed acquisition performance. Their PER results under strong jamming do not establish a delivery advantage for 2,500 hops/s. Therefore, choose the slower eligible rate, 1,250 hops/s, to reduce switching demands and sensitivity to dwell mismatch.

The recommendation is a practical choice among the tested settings, rather than proof of a universally optimal hop rate.

Conclusion

Increasing jammer occupancy raises BER in the tested channel, and interference creates an error floor at high E_b/N_0. Receiver dwell mismatch causes accumulated timing drift, with faster hopping showing greater sensitivity. FHSS has lower BER than a fixed link trapped on a jammed carrier, although a known clean fixed carrier performs better. The recommended configuration is eight channels over a nominal 195 kHz span at 1,250 hops/s, with 0.8 ms dwell time. Error correction, retransmission, or interference-aware channel selection would be needed to improve complete-packet delivery under the strong jammer used here.